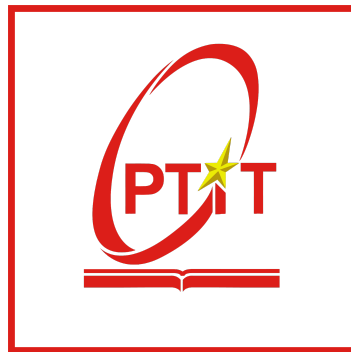


POST AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY
FACULTY OF INFORMATION TECHNOLOGY 1



FINAL REPORT
BUSINESS INTELLIGENCE
TOPIC: AIRBNB DATA ANALYSIS

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1. Introduction

The short-term rental market has experienced rapid growth over the past decade, with Airbnb emerging as the dominant global platform. New York City represents one of the largest and most competitive Airbnb markets in the world, attracting millions of visitors each year. Analysing this market provides valuable insights for hosts, investors, and urban policy planners.

This project builds a complete Business Intelligence (BI) solution to analyse the NYC Airbnb rental market. The system consists of a Data Warehouse designed using a Star Schema in PostgreSQL, an ETL pipeline implemented in Python, and an interactive Power BI dashboard for visual exploration.

Objectives:

- Analyse average rental prices across boroughs and neighbourhoods in New York City.
- Identify the most popular room types and their pricing characteristics.
- Measure booking rates as a proxy for market demand.
- Design and implement a scalable Data Warehouse to support BI queries.
- Build an interactive Power BI dashboard for visualisation and exploration.

Scope: The analysis covers 48,884 Airbnb listings across 5 boroughs and 221 neighbourhoods in New York City, based on the Inside Airbnb NYC 2019 dataset.

2. Dataset Description

2.1 Data Source

The dataset used in this project is the **Inside Airbnb New York City 2019** dataset, a publicly available collection of Airbnb listing data. The raw file is stored as AB_NYC_2019.csv.

2.2 Dataset Overview

Attribute	Value
Source	Inside Airbnb (insideairbnb.com)
City	New York City, USA
Year	2019
Raw Records	48,895 rows
Records after Cleaning	48,884 rows
Number of Boroughs	5
Number of Neighbourhoods	221
Number of Hosts	37,455

2.3 Key Fields

Field	Type	Description
id	BIGINT	Unique identifier for each listing
host_id	BIGINT	Unique identifier for each host
host_name	VARCHAR	Name of the host
neighbourhood_group	VARCHAR	Borough name (Manhattan, Brooklyn, etc.)
neighbourhood	VARCHAR	Specific neighbourhood within the borough
latitude / longitude	FLOAT	Geographic coordinates of the listing
room_type	VARCHAR	Room type (Entire home, Private room, Shared room)
price	INT	Nightly rental price in USD
minimum_nights	INT	Minimum number of nights per booking
number_of_reviews	INT	Total number of reviews received

reviews_per_month	FLOAT	Average monthly reviews
availability_365	INT	Number of days available per year (0–365)
calculated_host_listings_count	INT	Number of listings managed by the host

2.4 Data Quality Issues

The following issues were identified and resolved before loading into the Data Warehouse:

- **Missing values:** reviews_per_month nulls were filled with 0 (listings with no reviews). last_review nulls were filled with "No Review". host_name nulls were filled with "Unknown". name nulls were filled with "No Name".
- **Invalid records:** Listings with price = 0 or minimum_nights = 0 were removed as they represent erroneous or incomplete entries.
- **Duplicate records:** 11 duplicate rows were detected and removed using drop_duplicates(), resulting in 48,884 clean records.

3. Data Warehouse Design

3.1 Design Approach

The Data Warehouse is designed using a **Star Schema**, the most widely adopted structure for OLAP (Online Analytical Processing) workloads. The Star Schema consists of a central Fact Table connected to multiple Dimension Tables, enabling fast aggregation queries required by the Power BI dashboard.

3.2 Schema Overview

Table	Type	Description	Primary Key
fact_listing	Fact	Central table containing measurable metrics for each listing	listing_id
dim_location	Dimension	Geographic information: borough, neighbourhood, coordinates	location_id
dim_room_type	Dimension	Room type classification (3 types)	room_type_id
dim_host	Dimension	Host information and listing count	host_id

3.3 Fact Table — fact_listing

Column	Data Type	Description
listing_id	BIGINT PK	Unique listing identifier
host_id	BIGINT FK	Foreign key to dim_host

location_id	INT FK	Foreign key to dim_location
room_type_id	INT FK	Foreign key to dim_room_type
price	INT	Nightly price in USD — primary measure
minimum_nights	INT	Minimum booking duration
number_of_reviews	INT	Total reviews count
reviews_per_month	FLOAT	Average monthly reviews
availability_365	INT	Days available per year (used to compute booking rate)

3.4 Dimension Tables

dim_location (221 unique locations):

Column	Data Type	Description
location_id	INT PK	Surrogate key
neighbourhood_group	VARCHAR(100)	Borough name (5 values)
neighbourhood	VARCHAR(100)	Neighbourhood name (221 values)

latitude	FLOAT	Geographic latitude
longitude	FLOAT	Geographic longitude

dim_room_type (3 room types):

Column	Data Type	Description
room_type_id	INT PK	Surrogate key
room_type	VARCHAR(100)	Entire home/apt Private room Shared room

dim_host (37,455 unique hosts):

Column	Data Type	Description
host_id	BIGINT PK	Natural key from source data
host_name	VARCHAR(255)	Name of the host
calculated_host_listings_count	INT	Number of listings managed by this host

3.5 Relationships

- `fact_listing.location_id` → `dim_location.location_id` (Many-to-One)
- `fact_listing.room_type_id` → `dim_room_type.room_type_id` (Many-to-One)
- `fact_listing.host_id` → `dim_host.host_id` (Many-to-One)

4. ETL Process

The ETL (Extract, Transform, Load) pipeline is implemented entirely in Python and consists of four scripts located in the `etl/` directory.

4.1 Extract

Source: `dataset/raw/AB_NYC_2019.csv` — raw dataset containing 48,895 records and 16 columns. The file is loaded using the pandas library:

```
df = pd.read_csv("dataset/AB_NYC_2019.csv")
```

4.2 Transform — `clean_data.py`

The transformation step performs the following operations:

Step	Operation	Details
1	Handle null values	<code>reviews_per_month</code> → 0; <code>last_review</code> → "No Review"; <code>host_name</code> → "Unknown"; <code>name</code> → "No Name"
2	Remove duplicates	<code>drop_duplicates()</code> — removed 11 duplicate rows
3	Filter invalid price	Removed records where <code>price</code> = 0
4	Filter invalid nights	Removed records where <code>minimum_nights</code> = 0
5	Export clean data	Saved to <code>dataset/processed/airbnb_clean.csv</code> (48,884 rows)

4.3 Transform — `create_dimensions.py`

Dimension tables are extracted from the cleaned dataset using deduplication:

- **dim_location:** Unique combinations of `neighbourhood_group`, `neighbourhood`, `latitude`, `longitude` — surrogate key `location_id` assigned sequentially.

- **dim_room_type:** Unique room_type values — surrogate key room_type_id assigned sequentially.
- **dim_host:** Unique host records keyed on host_id (natural key from source). Exported with QUOTE_ALL to handle host names containing commas or quotes.

4.4 Transform — create_fact_table.py

The fact table is constructed by merging the cleaned dataset with dimension keys via merge() operations on shared columns (neighbourhood_group, neighbourhood, latitude, longitude for location; room_type for room type). Only foreign keys and measurable columns are retained: listing_id, host_id, location_id, room_type_id, price, minimum_nights, number_of_reviews, reviews_per_month, availability_365.

4.5 Load

All dimension and fact tables are loaded into **PostgreSQL** using the COPY command via SQLAlchemy and to_sql(). The database schema is defined in sql/create_tables.sql, which specifies primary keys, foreign keys, and appropriate data types.

Table	Records Loaded
dim_location	48,884 unique locations
dim_room_type	3
dim_host	37,455
fact_listing	48,884

5. Dashboard

5.1 Tool

The interactive dashboard is built using **Microsoft Power BI Desktop**, connected directly to the PostgreSQL Data Warehouse. The dashboard file is saved as dashboard/nyc_airbnb_dashboard.pbix.

5.2 Dashboard Components

Visual	Type	Objective Addressed
Total Listings, Avg Price, Avg Booking Rate, Neighbourhoods	KPI Cards (×4)	Overview metrics
Average Price by Borough	Horizontal Bar Chart	Rental price by area
Room Type Distribution	Donut Chart	Popular room types
Booking Rate by Borough	Horizontal Bar Chart	Booking demand by area
Listing Count by Price Range	Column Chart	Price distribution
Avg Price by Borough & Room Type	Matrix Table	Cross-analysis
Top Neighbourhoods by Listings	Horizontal Bar Chart	Most active areas

5.3 Filters / Slicers

- **Borough** — Dropdown slicer to filter by New York City borough.
- **Room Type** — Dropdown slicer to filter by room category.
- **Price Range (\$)** — Range slider slicer for minimum/maximum nightly price.

All slicers apply cross-filtering to all visuals on the page simultaneously, enabling dynamic exploration of the dataset.

5.4 DAX Measures

Avg Price = AVERAGE('public fact_listing'[price])

Booking Rate % = (1 - AVERAGE('public fact_listing'[availability_365]) / 365) * 100

Total Listings = COUNTROWS('public fact_listing')

6. Insight Analysis

6.1 Rental Price by Area

Manhattan dominates the NYC Airbnb market in terms of pricing, recording the highest average nightly price at **\$196.88** — significantly higher than the Bronx (\$87.58), the most affordable borough. Brooklyn ranks second at \$124.44, while Staten Island (\$114.81) and Queens (\$99.52) occupy the mid-range.

The overall average price across all 48,884 listings is **\$152.76**, reflecting the weight of Manhattan and Brooklyn which together account for over 85% of all listings.

Borough	Avg Price — Entire home/apt	Avg Price — Private room	Avg Price — Shared room
Manhattan	\$249.26	\$116.78	\$88.98
Brooklyn	\$178.35	\$76.55	\$50.77
Staten Island	\$173.85	\$62.29	\$57.44
Queens	\$147.05	\$71.76	\$69.02
Bronx	\$127.51	\$66.89	\$59.80

The cross-analysis by borough and room type reveals that Manhattan entire homes command the highest price at **\$249.26** per night — nearly double that of the Bronx (\$127.51). Interestingly, Staten Island entire homes (\$173.85) rank second, suggesting premium property characteristics despite lower overall listing volume.

Key insight: Manhattan's pricing premium is consistent across all room types. However, Brooklyn entire homes (\$178.35) offer a compelling alternative for hosts seeking high nightly rates with stronger booking demand.

6.2 Room Type Popularity

Entire homes/apartments are the most listed room type at **51.97%** of all listings (25,407), followed by private rooms at **45.66%** (22,319). Shared rooms represent a marginal **2.37%** (1,158).

The price gap is substantial: entire homes average **\$211.81** per night versus \$89.81 for private rooms and \$70.25 for shared rooms — a 136% premium for entire homes over private rooms. This reflects strong market segmentation: entire homes target tourists and groups seeking full privacy, while private rooms serve budget travellers and longer-stay guests.

Key insight: Despite a near-equal split in listing count between entire homes and private rooms, entire homes generate substantially higher revenue per booking. Hosts with entire properties in Manhattan can achieve up to \$249.26 per night on average.

6.3 Booking Rate Analysis

The average booking rate across all NYC listings is ****69.10%****, calculated as $(1 - \text{availability_365} / 365) \times 100$. This metric approximates the proportion of the year that a listing is occupied.

Borough	Booking Rate (%)
Brooklyn	~72% (highest)
Manhattan	~70%
Queens	~60%
Bronx	~55%

Staten Island	~45% (lowest)
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Brooklyn leads in booking rate despite not having the highest prices — indicating strong demand relative to supply and making it the most demand-efficient market in NYC. Staten Island records the lowest booking rate at approximately 45%, consistent with its limited public transport connectivity and lower tourist footfall.

Key insight: Brooklyn's combination of high booking rate (~72%) and moderate prices (\$124.44 average) makes it the strongest value proposition in the NYC market — for both guests seeking affordability and hosts seeking consistent occupancy.

6.4 Price Distribution

The price distribution is right-skewed, with listings heavily concentrated in the \$50–\$100 and \$100–\$150 bands:

Price Range	Listing Count (approx.)
< \$50	~6,500
\$50 – \$100	~17,000 (largest group)
\$100 – \$150	~10,000
\$150 – \$200	~6,500
\$200 – \$300	~5,000
\$300 – \$500	~2,500

> \$500	~1,000
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The \$50–\$100 range is the single largest price band, accounting for nearly one-third of all listings. Nearly **70% of all listings are priced below \$150 per night**, making the NYC Airbnb market relatively accessible to budget and mid-range travellers. The long tail above \$300 represents premium properties, primarily entire homes in Manhattan.

6.5 Top Neighbourhoods

The most active neighbourhoods by listing volume are:

Rank	Neighbourhood	Borough
1	Williamsburg	Brooklyn
2	Bedford-Stuyvesant	Brooklyn
3	Harlem	Manhattan
4	Bushwick	Brooklyn
5	Upper West Side	Manhattan
6	Hell's Kitchen	Manhattan
7	East Village	Manhattan

Williamsburg and Bedford-Stuyvesant in Brooklyn together represent the highest listing density of any neighbourhoods in NYC. Brooklyn accounts for 3 of the top 4

neighbourhoods, reflecting the borough's emergence as the primary alternative to Manhattan for both hosts and guests. Among Manhattan neighbourhoods, Harlem, Upper West Side, Hell's Kitchen, and East Village are the most active, offering a mix of budget and mid-range options relative to Midtown pricing.

6.6 Host Behaviour

Of the 37,455 unique hosts in the dataset, **5,154 hosts (13.8%)** manage more than one listing. This indicates a meaningful segment of professional or semi-professional hosts operating multiple properties — a trend associated with the growing commercialisation of the short-term rental market in NYC.

7. Conclusion

7.1 Summary

This project successfully demonstrates the design and implementation of a complete Business Intelligence pipeline applied to the NYC Airbnb rental market. The solution covers all stages from raw data acquisition through ETL processing, Data Warehouse design, and interactive dashboard delivery.

7.2 Key Findings

- Manhattan is the most expensive borough with an average nightly price of **\$196.88**, driven by premium entire-home listings in central neighbourhoods such as Midtown.
- Brooklyn offers the highest booking rate at **72.5%** while maintaining moderate pricing, positioning it as the most demand-efficient market in NYC.
- Entire homes/apartments dominate the market (52%) and command a **136% price premium** over private rooms, reflecting strong demand from tourists and groups.
- **70% of listings** are priced below \$150 per night, indicating the majority of the market targets budget-to-mid-range guests.
- **17,530 listings** showed zero availability, suggesting near-full occupancy for a significant segment of the market.

7.3 Technical Achievements

- Designed a normalised Star Schema Data Warehouse in PostgreSQL with 1 fact table and 3 dimension tables.
- Implemented a complete Python ETL pipeline: data cleaning (clean_data.py), dimension extraction (create_dimensions.py), fact table construction (create_fact_table.py), and database loading via SQLAlchemy.
- Built an interactive Power BI dashboard with 7 visualisations and 3 cross-filtering slicers, connected live to the PostgreSQL Data Warehouse.
- Created DAX measures for derived metrics: Avg Price, Booking Rate %, and Total Listings.

7.4 Limitations & Future Work

- The dataset represents a static snapshot from 2019. More recent data would provide more actionable insights, particularly given the significant impact of the COVID-19 pandemic on the short-term rental market.
- Booking rate is approximated from availability_365, not actual reservation data. True occupancy data from Airbnb is not publicly available.
- Future work could incorporate time-series analysis, sentiment analysis of reviews, and geospatial visualisation using latitude/longitude coordinates.

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